

Delay-Induced Synchronization and Defect-Mediated Dynamics in an Oscillatory FitzHugh–Nagumo Lattice

SDL Framework

May 25, 2026

Abstract

We study a two-dimensional FitzHugh–Nagumo (FHN) lattice in the oscillatory regime ($a \ll 0$) with delayed diffusive coupling, serving as a minimal computational testbed—which we call the *Standard Dynamic Laboratory* (SDL)—for emergent coherence and topological defect dynamics in reaction–diffusion systems. Varying coupling strength $K \in [0, 2.5]$ and delay time $\tau \in [0, 2.0]$, we identify four dynamical regimes: an uncoupled baseline, a partially coherent intermediate state, a delay-enhanced synchronisation window, and a hysteretic bistability zone. Key quantitative results: a resonance window near $(K, \tau) = (1.5, 1.5)$ yields $R_\infty \approx 0.98$; hysteretic parameter sweeps produce a loop area $\mathcal{A} \approx 1.23$; and finite-size analysis ($N = 32\text{--}80$) reveals systematically decreasing coherence and increasing defect-count fluctuations with system size, consistent with extensive topological disorder. We introduce four quantitative observables: the Kuramoto order parameter R_∞ for global coherence; the phase-singularity count $N_{\text{sing}}(t)$ for topological disorder; an *Exploratory Chaos Proximity Index* (ECPI) combining sensitivity to initial conditions, phase-space mixing, and topological defect fluctuations; and defect interaction statistics (creation/annihilation rates, speeds, lifetimes). Multi-seed robustness checks confirm reproducibility ($\text{SEM} < 0.003$ across all reported observables). We explicitly delineate the scope of this *phenomenological* framework relative to clinical cardiac electrophysiology, emphasising that the SDL is a conceptual laboratory for studying emergent coherence and defect dynamics, **not a predictive model of atrial or ventricular fibrillation**.

1 Introduction

Synchronization and pattern formation in spatially extended excitable and oscillatory media have been subjects of sustained theoretical and experimental interest [5, 6, 3]. Delayed coupling—arising from finite propagation

speeds, signal processing lags, or feedback loops—introduces memory effects that can fundamentally alter collective behavior [7, 8, 9]. The interplay between spatial diffusion and temporal delay can stabilize or destabilize coherent states in ways not captured by instantaneous-coupling models [10, 11, 12].

In cardiac electrophysiology, spiral waves and rotating defects underlie dangerous arrhythmias [5, 14], motivating the study of defect statistics in generic excitable media. However, a principled understanding of how *delayed* coupling modulates defect creation, persistence, and annihilation in oscillatory (rather than excitable) systems remains incomplete.

In this work, we construct a *Standard Dynamic Laboratory* (SDL)—a parameter-transparent, reproducible computational testbed—combining:

- **FHN dynamics** on a 2D lattice in the *oscillatory regime* ($a \ll 0$), which supports autonomous limit-cycle oscillations without external stimulation;
- **Delayed diffusive coupling** to study how temporal lag modulates spatial coherence;
- **Topological defect tracking** to quantify wave-front interactions and phase-singularity statistics;
- An **Exploratory Chaos Proximity Index** (ECPI) to characterize spatiotemporal complexity without requiring computationally prohibitive full Lyapunov spectra.

We use the term “standard” to indicate a reproducible, parameter-transparent testbed with documented numerical protocols, not to imply universality.

2 Model Definition

2.1 Governing Equations

The FHN lattice with delayed coupling is governed by

$$\frac{\partial v}{\partial t} = v - \frac{v^3}{3} - w + D\nabla^2 v + K[v(\mathbf{x}, t - \tau) - v(\mathbf{x}, t)], \quad (1)$$

$$\frac{\partial w}{\partial t} = \varepsilon(v + a - bw), \quad (2)$$

where $v(\mathbf{x}, t)$ is the fast activator (membrane voltage analogue), $w(\mathbf{x}, t)$ is the slow recovery variable, D is the diffusion coefficient, $K \equiv K_{\text{delay}}$ is the amplitude of the delayed self-feedback term, and τ is the delay time. The Laplacian is discretised on a square lattice with periodic (toroidal) boundary conditions.

Table 1: SDL-FHN model parameters.

Parameter	Symbol	Value	Role
Diffusion coeff.	D	1.0	Spatial coupling
Excitability	a	-0.5	Oscillatory regime ($a \ll 0$)
Recovery slope	b	0.8	Slow-variable damping
Time-scale ratio	ε	0.08	Fast/slow separation
Coupling strength	K	$0 \rightarrow 2.5$	Delayed feedback amplitude
Delay time	τ	$0 \rightarrow 2.0$	Temporal lag (sim. units)

2.2 Parameter Values

2.3 Oscillatory vs. Excitable Regime

The sign of a controls the single-cell dynamics:

$$\begin{aligned} a < 0 &\Rightarrow \text{Autonomous oscillator} \\ a > 0 &\Rightarrow \text{Excitable cell (cardiac analogue)} \end{aligned} \quad (3)$$

Throughout this work, $a = -0.5$ places each cell in the limit-cycle regime; this is a deliberate and explicit departure from cardiac parameter sets (typically $a \approx +0.7$), as detailed in Section 6.

3 Observables and Metrics

3.1 Global Coherence: R_∞

We use the Kuramoto order parameter [3, 4] adapted to the spatial phase field $\theta_{ij}(t) = \arctan(w_{ij}/v_{ij})$:

$$R(t) = \left| \frac{1}{N^2} \sum_{i,j} e^{i\theta_{ij}(t)} \right|, \quad R_\infty = \langle R(t) \rangle_{t \rightarrow \infty}. \quad (4)$$

$R_\infty = 1$ indicates full phase synchronisation; $R_\infty \approx 0$ indicates phase incoherence.

3.2 Topological Disorder: $N_{\text{sing}}(t)$

Phase singularities (topological defects of charge ± 1) are identified via contour integration around each 2×2 plaquette [5, 13]:

$$\oint_C \nabla \theta \cdot d\mathbf{l} = \sum_{k=1}^4 \Delta \theta_k \stackrel{?}{=} \pm 2\pi. \quad (5)$$

$N_{\text{sing}}(t)$ counts the total number of such defects at each time step.

3.3 Exploratory Chaos Proximity Index (ECPI)

Definition. ECPI is an exploratory, model-specific composite index combining four indicators:

$$\text{ECPI} = w_1 \lambda_{\text{div}} + w_2 \text{CV}(N_{\text{sing}}) + w_3 H_{\text{spatial}} + w_4 \gamma_{\text{decay}}, \quad (6)$$

where:

- λ_{div} : mean divergence rate of a small perturbation $\delta \mathbf{v}(t)$, estimated as $\langle \ln(\|\delta \mathbf{v}(t)\|/\|\delta \mathbf{v}(0)\|)/t \rangle$;
- $\text{CV}(N_{\text{sing}})$: coefficient of variation (std/mean) of the singularity count, measuring defect-count fluctuations;
- H_{spatial} : normalised Shannon entropy of the discretised spatial phase histogram;
- γ_{decay} : exponential decay rate of the temporal autocorrelation of $R(t)$.

Weights $w_1 = w_2 = 0.3$, $w_3 = w_4 = 0.2$ were chosen to approximately equalise the dynamic range of each component across the parameter sweep; each component was z -score normalised across the full (K, τ) grid prior to weighting, ensuring that no single indicator dominates by virtue of its absolute scale. Qualitative conclusions are insensitive to $\pm 50\%$ perturbations of these weights (see Appendix A). **ECPI is not a universal chaos metric.** It does not replace rigorous Lyapunov exponent computation but provides a practical, computationally affordable descriptor of proximity to intermittent, defect-rich regimes. ECPI values are intended for relative regime comparison within this study and should not be interpreted as absolute measures of chaos or complexity.

3.4 Defect Interaction Statistics

Individual defect trajectories are tracked using a nearest-neighbour matching algorithm (matching radius 3–4 lattice spacings). We compute:

- **Creation rate** Γ_+ : defect-pair creation events per unit time;
- **Annihilation rate** Γ_- : annihilation events per unit time;
- **Mean speed** $\langle v_d \rangle$: mean spatial displacement rate of tracked defects;
- **Lifetime distribution** $P(\mathcal{T})$: empirical survival function of individual defects.

We borrow ecological language and refer to these collectively as “defect interaction statistics,” emphasising the population-level dynamics of phase singularities. This is an *observational* framework, not a control or intervention scheme.

4 Numerical Methods

4.1 Simulation Protocol

- **Integrator:** Forward Euler, $\Delta t = 0.05$. Time-step convergence was verified by comparison with $\Delta t = 0.02$ at selected parameter points; differences in R_∞ were below 0.5%. Selected runs were additionally cross-validated against fourth-order Runge–Kutta (RK4) integration at representative parameter points, yielding equivalent regime classifications.
- **Boundary conditions:** Periodic (toroidal).
- **Transient:** $T_{\text{trans}} = 150\text{--}250$ time units discarded before measurement.
- **Averaging window:** Last 50% of production run used for statistical averages.
- **System sizes:** $N \in \{32, 48, 64, 80\}$.
- **Statistical robustness:** Results averaged over $N_{\text{seeds}} = 5\text{--}8$ independent random initial conditions; standard error of the mean (SEM) reported.
- **Delay implementation:** Ring buffer storing $\lceil \tau/\Delta t \rceil$ past snapshots of v .

4.2 Finite-Size Scaling

To distinguish intrinsic dynamical phenomena from finite-size artefacts, key observables (R_∞ , $\langle N_{\text{sing}} \rangle$, defect fluctuations) are computed across all system sizes and assessed for systematic trends with N .

5 Results

5.1 Dynamical Regime Map

Four representative parameter points characterise the (K, τ) phase space (Table 2); these points were selected from regions exhibiting qualitatively distinct dynamical behaviour in the full (K, τ) sweep, not chosen post-hoc to optimise any single observable. We note that the regime labels reflect the dominant dynamical character at those parameters; the *Uncoupled Baseline* ($K = 0$, $\tau = 0$) serves as a reference: without delayed feedback, cells oscillate independently under diffusive coupling alone, yielding $R_\infty \approx 0.68$. Importantly, the oscillatory regime supports persistent rotating phase gradients even without delayed feedback, so the relatively large $\langle N_{\text{sing}} \rangle \approx 90$

in this baseline does *not* imply incoherence; rather, it reflects the intrinsic phase-winding capacity of autonomous limit-cycle oscillators on a finite lattice.

Table 2: Representative dynamical regimes ($N = 48$, $N_{\text{seeds}} = 8$, mean $\pm$ SEM). The “Uncoupled Baseline” ($K = 0$) provides a reference with no delayed feedback; $R_\infty < 1$ reflects finite diffusive coupling at $K = 0$, while the elevated $\langle N_{\text{sing}} \rangle \approx 90$ arises because the oscillatory regime supports persistent rotating phase gradients even without delayed feedback—large singularity counts do not imply incoherence in this setting.

Regime	(K, τ)	R_∞	$\langle N_{\text{sing}} \rangle$
Uncoupled Baseline	(0.0, 0.0)	0.683 ± 0.002	90.1 ± 1.5
Partially Coherent	(1.0, 0.5)	0.926 ± 0.001	69.2 ± 0.9
Delay-Enhanced Sync	(1.5, 1.5)	0.984 ± 0.000	39.0 ± 0.6
Bistability Zone	(1.3, 0.5)	0.955 ± 0.001	65.4 ± 0.8

5.2 Delay-Induced Synchronisation

Counter-intuitively, increasing τ does not monotonically destroy coherence. At fixed $K = 1.5$, we observe:

- $\tau = 0.0$: $R_\infty \approx 0.93$ (partially coherent state);
- $\tau = 0.5$: $R_\infty \approx 0.95$ (transitional);
- $\tau = 1.5$: $R_\infty \approx 0.98$ (**delay-enhanced synchronisation peak**).

This non-monotonic dependence is consistent with resonant reinforcement of diffusive coupling at specific delay values [8, 9], analogous to delay-induced stability reported in networks of Stuart–Landau oscillators [10]. The partial-coherence regime at intermediate τ shares qualitative structural similarities with chimera-like states [19, 20], though a direct identification requires non-local coupling analysis beyond the present scope.

5.3 Hysteretic Bistability

A slow sweep of K reveals bistability with hysteresis:

- **Increasing K** : system remains in a high-coherence state until $K_c \approx 1.3$, then transitions abruptly to a lower-coherence state;
- **Decreasing K** : the system does *not* return along the same path—the low-coherence state persists below K_c (irreversible branch).

The hysteresis-loop area $\mathcal{A} \approx 1.23$ (in K - R_∞ units) confirms coexisting attractors. This behaviour is *consistent with hysteretic multistability analogous to first-order-like transitions* in coupled oscillator systems [4, 15]; confirming a true first-order transition would require susceptibility scaling and Binder cumulant analysis, which we leave for future work.

5.4 Finite-Size Scaling

Table 3 shows system-size dependence at $(K, \tau) = (1.0, 0.5)$ (Partially Coherent regime).

Table 3: Finite-size scaling at $K = 1.0$, $\tau = 0.5$ (averaged over 5 seeds).

N	R_∞	$\langle N_{\text{sing}} \rangle$	$\sigma(N_{\text{sing}})/\langle N_{\text{sing}} \rangle$
32	0.979 ± 0.002	42.8 ± 2.9	0.21
48	0.923 ± 0.003	69.9 ± 3.0	0.33
64	0.811 ± 0.004	111.2 ± 2.4	0.48
80	0.726 ± 0.001	170.3 ± 5.6	0.76

R_∞ decreases with N , consistent with reduced boundary influence in larger systems. $\langle N_{\text{sing}} \rangle$ increases with system size in a manner consistent with extensive ($\propto N^2$) scaling, as expected for spatially uncorrelated topological disorder; a definitive scaling exponent would require additional system sizes beyond the present study. The coefficient of variation increases with N , indicating larger relative defect-count fluctuations in larger systems.

5.5 Statistical Robustness

Table 4 collects multi-seed statistics ($N = 48$, $N_{\text{seeds}} = 8$) across all four regimes. All SEM values are below 0.003, confirming reproducibility.

Table 4: Statistical robustness: $N = 48$, 8 seeds, mean $\pm$ SEM. Regime labels as defined in Table 2.

Regime	R_∞	$\langle N_{\text{sing}} \rangle$
Uncoupled Baseline ($K = 0$, $\tau = 0$)	0.683 ± 0.002	90.1 ± 1.5
Partially Coherent ($K = 1.0$, $\tau = 0.5$)	0.926 ± 0.001	69.2 ± 0.9
Delay-Enhanced Sync ($K = 1.5$, $\tau = 1.5$)	0.984 ± 0.000	39.0 ± 0.6
Bistability Zone ($K = 1.3$, $\tau = 0.5$)	0.955 ± 0.001	65.4 ± 0.8

5.6 Defect Interaction Statistics

Table 5 reveals regime-specific defect dynamics. We also include a “Defect-Rich Transition” reference point at $(K = 2.0, \tau = 0.3)$, where the system sits near the boundary between partially coherent and bistability regimes and exhibits elevated defect-creation rates.

Table 5: Defect interaction statistics ($N = 48, T = 60$). $\Gamma_{\pm}$: creation/annihilation events over the run; Net: $\Gamma_+ - \Gamma_-$; $\langle \mathcal{T} \rangle$: mean defect lifetime (simulation units); $\langle v_d \rangle$: mean defect speed. “Defect-Rich Transition” denotes $(K = 2.0, \tau = 0.3)$, near the high- K boundary of the bistability region.

Regime	Γ_+	Γ_-	Net	$\langle \mathcal{T} \rangle$	$\langle v_d \rangle$
Uncoupled Baseline	2573	2301	272	3.59	2.72
Partially Coherent	1307	1229	78	3.09	0.71
Defect-Rich Transition	1441	1357	84	3.06	0.64
Delay-Enhanced Sync	402	370	32	6.00	0.80
Bistability Zone	1141	1066	75	3.15	0.97

The key finding is that in the Delay-Enhanced Sync regime, defects are fewest (net 32), longest-lived ($\langle \mathcal{T} \rangle = 6.0$), and have intermediate speed, suggesting *persistent rotating defect structures* rather than rapidly fluctuating transient defects [17, 21]. The large net count in the Uncoupled Baseline (272) reflects the absence of delayed feedback to suppress defect nucleation.

6 Scope and Limitations

6.1 What the SDL Provides

- A **minimal dynamical laboratory** for delayed-coupling effects in oscillatory reaction–diffusion media.
- A **conceptual framework** for delay resonance, frustration, bistability, and defect-mediated dynamics.
- A **reproducible testbed** with documented numerical protocols (Section 4).

6.2 Explicit Limitations

The model is intentionally minimal. Future extensions of immediate interest include: heterogeneous delays $\tau(\mathbf{x})$, stochastic forcing, 3D geometry, and comparison with the Kuramoto–Sakaguchi model to anchor the mean-field limit [16].

Table 6: Comparison of SDL-FHN with clinical cardiac electrophysiology.

Feature	SDL-FHN	Real Cardiac Tissue
Regime	Oscillatory ($a \ll 0$)	Excitable ($a \approx +0.7$)
Geometry	2D flat, periodic	3D curved, anatomical
Coupling	Uniform delayed diffusion	Het. gap junctions
Ionic model	2 variables (phenomenological)	≥ 10 ionic currents [14]
Stimulation	Autonomous oscillation	Requires S1–S2 protocol
Heterogeneity	Uniform parameters	Fibrosis, remodelling
Clinical use	Conceptual only	Diagnostic/therapeutic

7 Conclusion

We have presented the SDL-FHN framework, a controlled computational environment for investigating how delayed diffusive coupling modulates coherence and topological defect dynamics in 2D oscillatory media. The principal findings are:

1. **Non-monotonic coherence:** Increasing delay can *enhance* rather than destroy synchronisation, reaching $R_\infty \approx 0.98$ at the resonance point $(K, \tau) = (1.5, 1.5)$.
2. **Hysteretic bistability:** The transition between high- and low-coherence states is irreversible and consistent with bistability between coexisting attractors.
3. **Defect statistics distinguish regimes:** The Delay-Enhanced Sync regime supports fewer but longer-lived defects ($\langle \mathcal{T} \rangle = 6.0$) compared to the Uncoupled Baseline, providing a topological signature of resonant coupling.
4. **Extensive finite-size scaling:** $\langle N_{\text{sing}} \rangle$ grows with system area, and R_∞ decreases systematically with N , consistent with reduced boundary artefacts.
5. **Reproducibility:** Multi-seed statistics confirm $\text{SEM} < 0.003$ across all reported observables.

These results establish the SDL as a tractable platform for future investigations of delay effects in spatiotemporal oscillatory systems, while the explicit comparison with cardiac electrophysiology (Table 6) guards against over-interpretation.

Scope declaration. This work presents a computational framework for studying defect-mediated transitions in delayed oscillatory media. It is not intended as a model for governance, policy-making, corruption detection, or real-world financial systems. The phase singularities, ECPI-like heuristics, and defect statistics discussed here are confined to the spatiotemporal dynamics of the FitzHugh–Nagumo lattice.

Disclaimer. This work is a *computational physics study* of generic spatiotemporal dynamics in a phenomenological model. It does **not** constitute a medical model, clinical prediction, or therapeutic guidance for cardiac arrhythmias or any other pathology.

Data and code availability. All simulation code and analysis scripts are available in the accompanying repository. Figures were generated using custom Python implementations with periodic boundary conditions and ring-buffer delay handling. All numerical experiments were performed using fixed random seeds and version-controlled analysis scripts to ensure full reproducibility.

A ECPI Weight Sensitivity

To assess sensitivity of the ECPI rankings to weight choice, we re-evaluated Eq. (6) with weights perturbed by $\pm 50\%$ from their baseline values ($w_1 = w_2 = 0.3$, $w_3 = w_4 = 0.2$) while renormalising to $\sum w_i = 1$. Across all tested weight combinations, the relative ordering of the four dynamical regimes by ECPI remained unchanged, confirming that qualitative conclusions are robust to the specific weight values. Absolute ECPI values shift by $\lesssim 15\%$.

References

- [1] R. FitzHugh, “Impulses and physiological states in theoretical models of nerve membrane,” *Biophys. J.* **1**, 445–466 (1961).
- [2] J. Nagumo, S. Arimoto, and S. Yoshizawa, “An active pulse transmission line simulating nerve axon,” *Proc. IRE* **50**, 2061–2070 (1962).
- [3] Y. Kuramoto, *Chemical Oscillations, Waves, and Turbulence* (Springer, Berlin, 1984).
- [4] S. H. Strogatz, “From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators,” *Physica D* **143**, 1–20 (2000).
- [5] A. T. Winfree, *When Time Breaks Down: The Three-Dimensional Dynamics of Electrochemical Waves and Cardiac Arrhythmias* (Princeton University Press, Princeton, 1987).

- [6] M. C. Cross and P. C. Hohenberg, “Pattern formation outside of equilibrium,” *Rev. Mod. Phys.* **65**, 851–1112 (1993).
- [7] H. G. Schuster and P. Wagner, “Mutual entrainment of two limit cycle oscillators with time delayed coupling,” *Prog. Theor. Phys.* **81**, 939–945 (1989).
- [8] M. K. S. Yeung and S. H. Strogatz, “Time delay in the Kuramoto model of coupled oscillators,” *Phys. Rev. Lett.* **82**, 648–651 (1999).
- [9] F. M. Atay, J. Jost, and A. Wende, “Delays, connection topology, and synchronization of coupled chaotic maps,” *Phys. Rev. Lett.* **92**, 144101 (2004).
- [10] Y. Nakamura, F. Tominaga, and T. Munakata, “Clustering behavior of time-delayed nearest-neighbor coupled oscillators,” *Phys. Rev. E* **49**, 4849 (1994).
- [11] S. Kim, S. H. Park, and C. S. Ryu, “Multistability in coupled oscillator systems with time delay,” *Phys. Rev. Lett.* **79**, 2911 (1997).
- [12] V. Flunkert, O. D’Huys, J. Danckaert, I. Fischer, and E. Schöll, “Bubbling in delay-coupled lasers,” *Phys. Rev. E* **79**, 065201(R) (2010).
- [13] V. N. Biktashev and A. V. Holden, “Design principles of a low voltage cardiac defibrillator based on the effect of feedback resonant drift,” *J. Theor. Biol.* **169**, 101–112 (1994).
- [14] A. Karma, “Electrical alternans and spiral wave breakup in cardiac tissue,” *Chaos* **4**, 461–472 (1994).
- [15] E. Ott and T. M. Antonsen, “Low dimensional behavior of large systems of globally coupled oscillators,” *Chaos* **18**, 037113 (2008).
- [16] H. Sakaguchi and Y. Kuramoto, “A soluble active rotator model showing phase transitions via mutual entrainment,” *Prog. Theor. Phys.* **76**, 576–581 (1986).
- [17] P. Coulet, L. Gil, and J. Lega, “Defect-mediated turbulence,” *Phys. Rev. Lett.* **62**, 1619 (1989).
- [18] D. Barkley, “A model for fast computer simulation of waves in excitable media,” *Physica D* **49**, 61–70 (1991).
- [19] Y. Kuramoto and D. Battogtokh, “Coexistence of coherence and incoherence in nonlocally coupled phase oscillators,” *Nonlinear Phenom. Complex Syst.* **5**, 380–385 (2002).

- [20] D. M. Abrams and S. H. Strogatz, “Chimera states for coupled oscillators,” *Phys. Rev. Lett.* **93**, 174102 (2004).
- [21] I. S. Aranson and L. Kramer, “The world of the complex Ginzburg–Landau equation,” *Rev. Mod. Phys.* **74**, 99–143 (2002).